IMS Bearing Run-to-Failure: Health, Fault and RUL

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On three NASA/IMS run-to-failure sets, an anomaly detector and a Random Forest find damage onset, XGBoost predicts remaining life with about 9% of life average error, and fault diagnosis works for only two of the three bearings.

Method

- **Data:** three IMS sets (case 1 bearing 3, case 2 bearing 1, case 3 bearing 3) with 21 per-file features: RMS, peak, kurtosis, skew, crest factor, four band energies, and envelope peaks at BPFO, BPFI and BSF.
- **Damage label:** the first 30% of each run is healthy. After that a file is damaged once the 5-file rolling-median RMS exceeds the healthy median plus 3 times the IQR, and it stays damaged. Onset falls at 85.7% of life in case 1, 54.3% in case 2 and 82.1% in case 3.
- **Life-left target:** 100% until onset, then a straight line down to 0% at the last file.
- **Normalisation:** every feature is divided by that set's own median over its first 20% of files.
- **Testing:** leave-one-set-out. Train on two sets, test on the third, repeat three times. No random splits.
- **Models:** Isolation Forest fitted on the first 20% (alarm after 5 flagged files in a row), Random Forest for health, a BPFO/BPFI ratio rule and logistic regression for fault, Ridge, Random Forest and XGBoost for life left.

Results

Task 1a: anomaly alarm vs labelled onset (negative = alarm came first)

Set	Onset (file)	Alarm (file)	Lag (files)	Lag (% of life)
case 1	1847	1788	-59	-2.7
case 2	534	639	+105	+10.7
case 3	5191	6015	+824	+13.0

Task 1b: Random Forest, damaged vs healthy (leave-one-set-out, all sets pooled)

Feature set	Accuracy	Precision (damaged)	Recall (damaged)	F1 (damaged)
With RMS (circular, the label comes from RMS)	0.925	0.933	0.673	0.782
Without RMS-like features	0.822	0.570	0.437	0.495

Task 2: which ring broke (accuracy on damaged files; the rule says BPFO when the healthy-normalised BPFO/BPFI ratio is above 1)

Set	True fault	Rule	Linear model
case 1	BPFI	1.000	undefined (no BPFI left in training)
case 2	BPFO	1.000	0.673
case 3	BPFO	0.147	0.814

Task 3: life-left error (MAE in percentage points of life, leave-one-set-out)

Model	Mean MAE, all files	Mean MAE, damaged files only
Ridge	9.93	33.10
Random Forest	8.49	20.53
XGBoost	9.03	19.29

Limitations

- **No ground-truth onset.** The damage label is a rule on RMS, not a measured event. In case 1 the detector alarms slightly before it, and in case 3 the anomaly score starts drifting well before the label.
- **Circular health score.** The label is built from RMS, so the 0.925 accuracy with RMS partly measures how well the model reproduces its own label. Use the 0.822 figure without RMS-like features as the honest one.
- **Fault diagnosis rests on one inner-race bearing.** Only case 1 is BPFI, so the linear model cannot be trained when case 1 is held out. Case 3's outer-race fault barely shows in the BPFO and BPFI envelope peaks, and the rule gets only 14.7% right there. Trying five other filter bands did not fix it: no band worked for all three bearings.

- **Life left is weak once damage starts.** The best model is off by 19.3 points of life on damaged files, against 9.0 on all files, because healthy files are easy.
- **Only three bearings.** Every number is an average over three folds, so small differences between models are not reliable.

Reproduce

```
python ims_cli.py extract IMS               # builds
case{1,2,3}_features_v2.csv, skips any that already exist
python ims_cli.py extract IMS --force       # rebuild them
python ims_cli.py run all --data . --plots plots
```

The same steps are in `02_RUL_and_Fault_Classification_v3.ipynb`, which reads the three `_v2.csv` files.